Muon Neutrino Charged Current Single Pion Cross Section on Argon using Run 1 NuMI Data

Internal Note — Version 0.2 (DRAFT)

> ***DRAFT STATUS — READ FIRST***
>
> *This note mirrors the structure of the BNB CC1π internal note (`BNBInternalNoteCC1pi_V0.91.pdf`, J. P. Detje, 4 Feb 2025) for the NuMI Run 1 FHC measurement.*
>
> *Sections 1–5 describe the analysis* ***as it is actually configured in the code today*** *and every number in them was read from the source or configs. Section 6 (fake-data tests) now carries* ***real results*** *from the re-run chain: the GENIE closure χ^2 for all five observables and the four-generator comparison on the corrected flux. Section 7 (real-data results) is* ***still empty*** *— the real beam-on sample has not been restored and the chain has not been run on data (see §2.2 and §7).*
>
> *The single most important change since v0.1 is the* ***flux-normalisation correction*** *(§2.4): the NuMI flux constant was ~3.48× too large, which had made every extracted cross section ~3.5× too small. Every number in §6 uses the corrected flux. Nothing in §7 should be quoted as a data result.*

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1 Overview

This note describes the extraction of the νμ charged-current single charged pion cross section on argon using MicroBooNE Run 1 FHC data from the NuMI beam. It is the NuMI counterpart of the BNB analysis of the same final state, and is intended to be read alongside that note, which contains fuller descriptions of the selection variables and the boosted decision trees.

The measurement is performed with the common `xsec_analyzer` framework: event selection via the `CC1mu1piXp` selection class, universe construction via `univmake`, and unfolding via the Wiener-SVD implementation in `CrossSectionExtractor`. Section 2 describes the software and samples, Section 3 the signal definition and selection, Section 4 the uncertainty treatment, Section 5 the cross-section extraction and unfolding, Section 6 the fake-data tests, and Section 7 the results.

The principal differences from the BNB analysis are:

	BNB analysis	**This analysis**
Beam	BNB	NuMI FHC
Runs	1–5	1 only
Flux systematic	`weight_flux_all`	`weight_ppfx_all` (PPFX)
Beamline geometry systematic	n/a	NuMI-specific — **not currently included**, see §4.6
Proton multiplicity	—	Xp (any number, no threshold)
Unfolding	Wiener-SVD	Wiener-SVD (D'Agostini available as a cross-check)

2 Software and Samples

2.1 Framework

The analysis runs entirely within `xsec_analyzer`. The chain is:

1. `ProcessNTuples` — applies the `CC1mu1piXp` selection to raw PeLEE ntuples and writes `stv_tree` output (`run_process_ntuples.sh`).
2. `univmake` — builds systematic universe histograms from a bin config (`run_universe_maker.sh`).
3. `Unfolder` / `UnfolderNuMI` — extracts and unfolds the cross section (`run_unfolder.sh`).

2.2 Samples

Run 1 FHC, as configured in `configs/file_properties_numi.txt`:

Sample type	**Purpose**
`onBNB`	beam-on data (see warning below)
`extBNB`	beam-off (EXT), 3 821 593 triggers
`numuMC`	NuMI overlay CV
`dirtMC`	dirt overlay
`detVarCV` + 8 variations	detector systematics

> ***The `onBNB` slot currently holds GENIE fake data.*** *It is the detector-variation central-value sample (a GENIE G18_10a production), symlinked as `xsec-ana-genie_detvarCV_fakedata_run1_fhc.root` and entered with its native POT (7.631×10^{20}) and beam-on-equivalent triggers (18 153 256). This sample is statistically **independent** of the numuMC overlay that builds the response matrix, so it is a genuine same-generator closure rather than a trivial self-closure. It replaced the earlier NuWro overlay fake data (`xsec-ana-numi_nuwro_overlay_pion_ntuples_run1_fhc.roo` 6.65×10^{20} POT), which carried a real $\cos\theta\_\mu$* shape *difference versus any standalone NuWro (§6.4).*
>
> *The real beam-on sample (`xsec-ana-neutrinoselection_filt_run1_beamon_beamgood.root`, 3.283×10^{20} POT, 7 809 962 triggers) remains **commented out** at `configs/file_properties_numi.txt.`*

*The analysis is therefore configured as a **fake-data study**, not a real-data measurement. Section 7 cannot be filled until this is swapped back and the chain re-run on data.*

2.3 Detector variation samples

Eight variations are used: `detVarLYdown`, `detVarLYrayl`, `detVarRecomb2`, `detVarSCE`, `detVarWMAngleXZ`, `detVarWMAngleYZ`, `detVarWMX`, `detVarWMYZ`.

The BNB note additionally lists `detVarLYatten`; it is not present in the NuMI sample set here. Like the BNB analysis, the WireMod dE/dx variation is not used, in favour of the recombination variation.

2.4 Corrections applied since the previous iteration

The following were found and fixed during a code review of this analysis. All of them affect the numbers a run would produce, which is why no results from before them should be carried forward:

- **Flux normalisation (dominant correction).** The integrated NuMI flux constant in the cross-section conversion factor was 2.372512×10^{-9} νμ/cm²/POT, taken from the superseded `uboone_numi_flux_histograms.root` (which included an unphysical ~25–30 MeV artifact spike). The authoritative value from the flux-file author, using the new GEANT4 flux integrated above 60 MeV, is 4.43515×10^{-10} (ν_μ) + 2.37644×10^{-10} ($\bar{\nu}_\mu$) = 6.81159×10^{-10} νμ/cm²/POT — a factor **3.48 smaller**. Because the cross section divides by the flux, the old constant made every extracted cross section ~3.5× too *small*. The correction was confirmed four independent ways: standalone NuWro (2.7× low) and GENIE (3.2× low) run on the old flux; the flux author's numbers (3.48×); and a GENIE closure in which the framework tune scaled by 3.48× matches a standalone GENIE on the corrected flux to 3%. Fixed in `FiducialVolume.hh` (`integrated_numu_flux_in_FV`). The flux is the active-volume-averaged flux used as an approximation for the fiducial-volume flux — a documented ~few-% residual; no fiducial-volume correction is applied (the target count is done separately in the same conversion factor).
- **Muon momentum estimator.** The estimator combining range (contained) and MCS (uncontained) was computed but never written to the output tree, so the unfolding binned on MCS alone. It is now branched as `candidate_muon_mom_reco` and `ccpi_pmu_bin_config.txt` uses it.
- **Pion momentum estimator.** Reco pion momentum was a proton-hypothesis range *kinetic energy* binned against a true *momentum* on identical bin edges. It is now the muon-hypothesis range momentum.
- **Per-category systematics.** Six `MCFullCorrCategory` entries evaluated to identically zero and have been removed; see §4.6.
- **Background true bins.** `MakeConfig` had stopped emitting them, so any regenerated bin config would have had no background subtraction at all.
- **Diagnostic histograms** are now filled with the CV weight (`spline_weight_ × tuned_cv_weight_`).

3 Signal and Selection

3.1 Signal definition

An event is signal if, in truth:

- the neutrino vertex is inside the fiducial volume: 10 < x < 246 cm, −101 < y < 101 cm, 10 < z < 986 cm;
- the interaction is charged-current;
- the neutrino is νμ (`|mc_nu_pdg| == 14`, see §8.3);
- there is exactly one muon above threshold;
- there is exactly one charged pion;
- there are no neutral pions, no kaons and no heavier mesons;
- any number of protons is allowed, with no momentum threshold (the ”Xp” in `CC1mu1piXp`);

and the event falls in the measured phase space:

- p_μ > 0.15 GeV/c
- p_π > 0.175 GeV/c
- θ_μπ < 2.6 rad

The phase-space cuts are applied identically in `define_signal()` and `categorize_event()`, so signal and category assignment cannot disagree.

3.2 Binning

Five differential observables are measured. Bin edges as configured:

Observable	**Config**	**Bins (true/reco)**	**Edges**
p_μ	`ccpi_pmu_bin_config.txt`	23/22	0.15, 0.20, 0.30 … 2.00, 2.30, 2.60, 3.00 GeV/c
p_π	`ccpi_ppi_bin_config.txt`	6/5	0.175, 0.20, 0.30, 0.40, 0.50, 1.00 GeV/c
cos θ_μ	`ccpi_costhmu_bin_config.txt`	13/12	−1.0, −0.5, 0.0, 0.2, 0.4, 0.55, 0.65, 0.75, 0.82, 0.88, 0.93, 0.97, 1.0
cos θ_π	`ccpi_costhpi_bin_config.txt`	13/12	−1.0, −0.7, −0.4, −0.2, 0.0, 0.2, 0.35, 0.5, 0.65, 0.78, 0.88, 0.95, 1.0
θ_μπ	`ccpi_thmupi_bin_config.txt`	10/9	0.0 … 2.513, 2.600 rad

Each true-bin count is one greater than the reco count: the extra true bin is the background bin, defined by inverting the signal definition (`!CC1mu1piXp_MC_Signal`). All configs are single-block with no sideband reco bins.

3.3 Particle identification

Three BDTs are used, trained separately and applied via `TMVA::Reader`:

Reader	**Weights (`booster_decision_tree/`)**	**Role**
`tmvaReader`	`dataset_MIP_BDT_no_len`	MIP identification
`tmvaReader_mu`	`dataset_muon_BDT`	muon — **booked but never evaluated**, see §8.4
`tmvaReader_pi`	`dataset_pion_BDT_no_len`	pion

Input variables: the three-plane Bragg likelihood ratios (`trk_bragg_p_v`, `trk_bragg_mu_v`, `trk_bragg_mip_v`), the LLR PID score, the track score, track length, and the SCE-corrected track end position. The pion reader omits track length.

3.4 Event selection cuts

Applied in order; an event passes only if all are satisfied:

1. software trigger
2. reconstructed vertex in the fiducial volume
3. topological score
4. muon candidate is track-like
5. pion candidate is contained
6. muon candidate not in a detector gap
7. pion candidate not in a detector gap
8. shower cut
9. muon–pion opening angle cut
10. final track-multiplicity cut

3.5 Selection efficiency and purity

The CV-weighted selection performance, measured on the numuMC overlay (cos θ_μ binning; efficiency and purity are properties of the selection and are essentially observable-independent):

Quantity	Value
Total selected (reco)	~3.71×10^3 events
Selected signal	~2.21×10^3 events
All true signal (denominator)	~1.42×10^4 events
Purity	**0.60**
Efficiency	**0.16**

The selection- and unfolding-stage diagnostics produced by `UnfolderNuMI` for cos θ_μ are shown in the figures below (the same four stages are produced for every observable):

- **Reco spectrum** — the selected reconstructed distribution, with the MC split into signal (CC1μ1πXp) and MC+EXT background, compared to the data (here the GENIE fake data of §2.2).
- **Background subtraction** — the data after subtracting the estimated background, with the MC signal overlaid; this is the input to the unfolding.
- **Smearceptance (response) matrix** — the reco-vs-true migration matrix that the Wiener-SVD unfolding inverts.
- **Efficiency** — the selection efficiency versus the true observable.

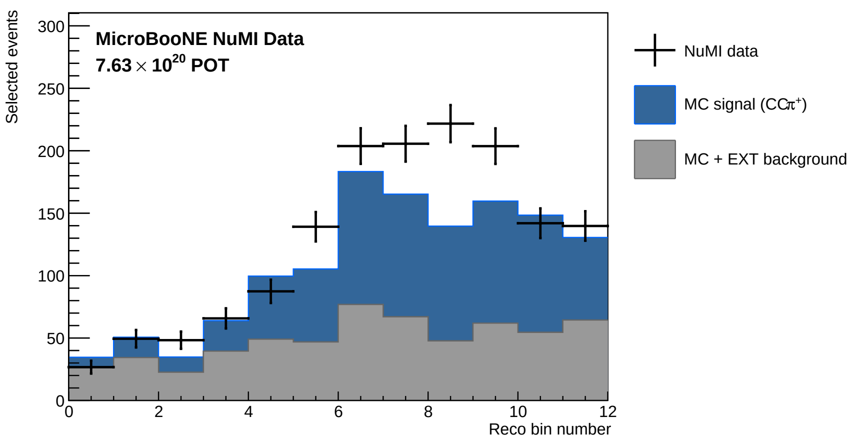

Figure 1: Selected reco $\cos\theta_\mu$ spectrum: data (GENIE fake data) vs MC signal + MC/EXT background.

4 Uncertainties

Configured in `configs/ccpi_systcalc_numi.conf`. The total is

```
total = detVar_total + flux_total + reint + xsec_total
        + POT + numTargets + SimulationStats + DataStats
```

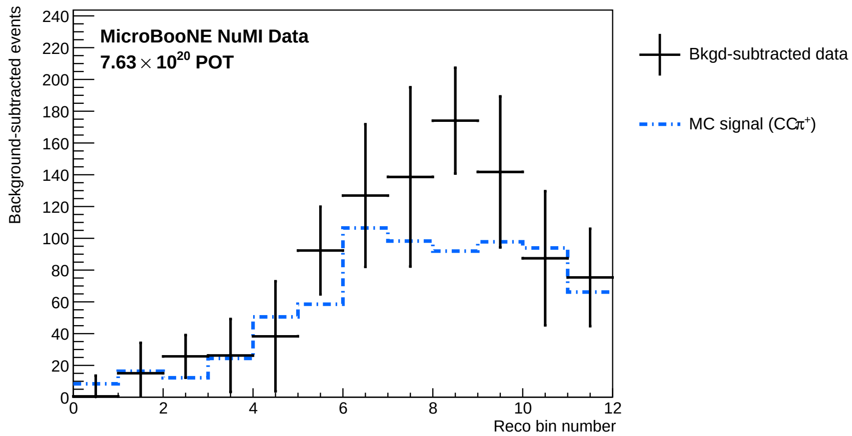

Figure 2: Background-subtracted reco spectrum with the MC signal overlaid (input to unfolding).

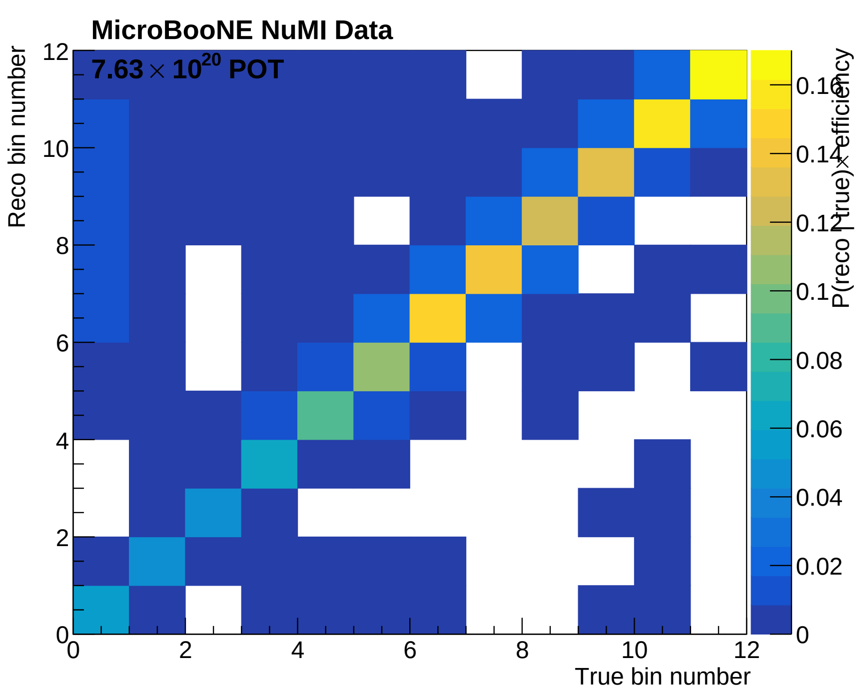

Figure 3: Smearceptance (response) matrix, reco vs true $\cos\theta_\mu$.

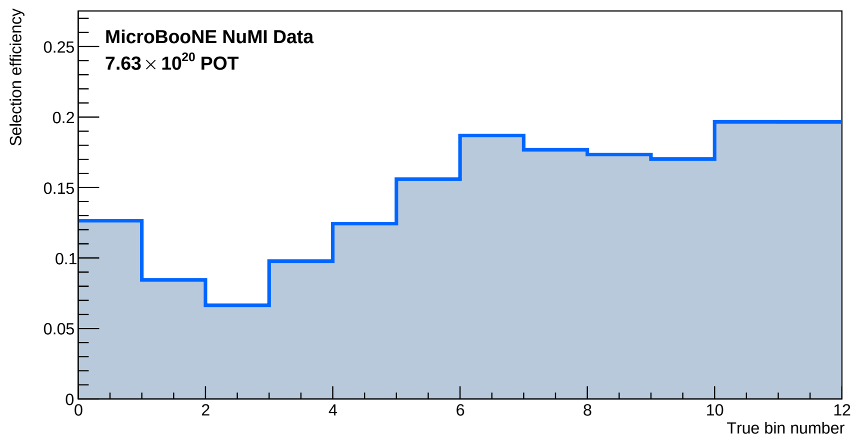

Figure 4: Selection efficiency vs true $\cos\theta_\mu$.

4.1 Flux

PPFX hadron-production multisims via `weight_ppfx_all`. A 2% fully-correlated normalisation uncertainty is applied for POT (`POT MCFullCorr 0.02`).

4.2 Interaction

GENIE multisims (`weight_All_UBGenie`) plus nine unisim knobs (`AxFFCCQEshape`, `DecayAngMEC`, `NormCCCOH`, `NormNCCOH`, `RPA_CCQE`, `ThetaDelta2NRad`, `Theta_Delta2Npi`, `VecFFCCQEshape`, `XSecShape_CCMEC`) and the two second-class-current form factors (`xsr_scc_Fa3_SCC`, `xsr_scc_Fv3_SCC`), matching the BNB treatment.

4.3 Reinteraction

`weight_reint_all` multisims.

4.4 Detector

The eight detector variation samples of §2.3 (plus the detVar central value), evaluated as `DV` type. The figures at the end of this section overlay all nine samples for every observable, POT-scaled to the CV exposure, with the reconstructed spectrum in the upper pad and the true signal distribution in the lower pad. Two features are visible and expected:

- the **reco** distributions show a clear sample-to-sample spread — this is the detector systematic;
- the **true** distributions all lie on top of the CV — the variations re-simulate the detector response on fixed generator events, so they change the reconstruction but not the underlying truth.

The reco spread, propagated through the response matrix, is the detector contribution to the uncertainty budget. The samples are the Run-3b detVar set applied globally (§4.6).

4.5 Target

A 1% normalisation uncertainty on the number of argon nuclei in the fiducial volume (`numTargets MCFullCorr 0.01`), as in the BNB analysis.

4.6 Known gaps in the uncertainty budget

Two components are **not** included, and the total above is correspondingly under-estimated:

NuMI beamline geometry. The beamline variation weights (`weight_Horn_2kA`, `weight_Horn1_x_3mm`, `weight_Beam_spot_*`, etc.) are absent from the processed ntuples. `instructions_numi.txt` documents `AddBeamlineGeometryWeights` as the tool that injects them, but no driver script invokes it, and the required histogram file (`NuMI_Geometry_Weights_Histograms.root`, canonical copy at `/exp/uboone/data/users/pgreen/NuMIFlux/NewFluxFiles/`) is not available locally. This systematic has no BNB counterpart and must be restored before the measurement is complete.

Background normalisation by category, and dirt normalisation. Six `MCFullCorrCategory` entries were removed because they evaluated to identically zero — they matched true bins by exact string equality against `"category == N"` while the bin configs declare a single `"!CC1mu1piXp_MC_Signal"` background bin. Their category indices were also inherited from the NuMI CC1e analysis and did not correspond to this selection's categories. Removing them changed no number (verified: 152 histograms bit-identical). Full detail in `configs/ccpi_systcalc_numi.README.md`.

Note the BNB uncertainty breakdown (Fig. 43–44 of that note) comprises `EXTstats, MCstats, POT, detVar_total, flux, numTargets, reint, xsec_total` — i.e. the same set retained here, with no per-category or dirt term.

5 Cross-section Extraction

The flux-integrated differential cross section in truth bin α is obtained by subtracting background from the measured event rate, unfolding to truth space, and dividing by the integrated flux Φ, the number of argon targets T, and the bin width:

```
dσ/dx |α  =  (1 / (T · Φ · Δx_α)) · Σ_a U_αa (D_a − B_a)
```

where `D_a` is the measured rate in reco bin a, `B_a` the estimated background, and `U_αa` the unfolding matrix, which absorbs both the reco→truth mapping and the efficiency.

Targets are counted as **argon nuclei** in the fiducial volume, so the result is per-nucleus, not per-nucleon — matching the BNB convention ($T = 1.03 \times 10^{30}$ nuclei there). For NuMI the fiducial volume additionally excludes the dead region $675.1 < z < 775.1$ cm.

Results are quoted in units of **10^{-38} cm^2 / Ar**, differential in the observable (so 10^{-38} cm^2 / GeV / Ar for momenta, 10^{-38} cm^2 / rad / Ar for angles).

Both the NuMI unit factor (now `1e38`) and the bin-width division in `Unfolder.C` were corrected to this convention; `Unfolder` and `UnfolderNuMI` now agree bin-for-bin (§8.1).

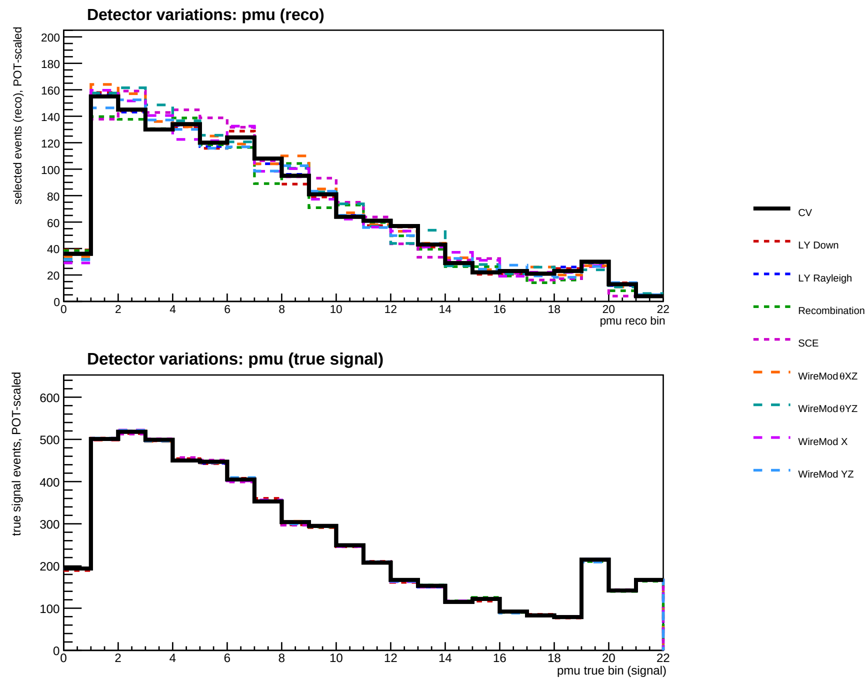

Figure 5: Detector variations, p_μ: reco spectrum (top) and true signal (bottom), all nine samples POT-scaled to the CV.

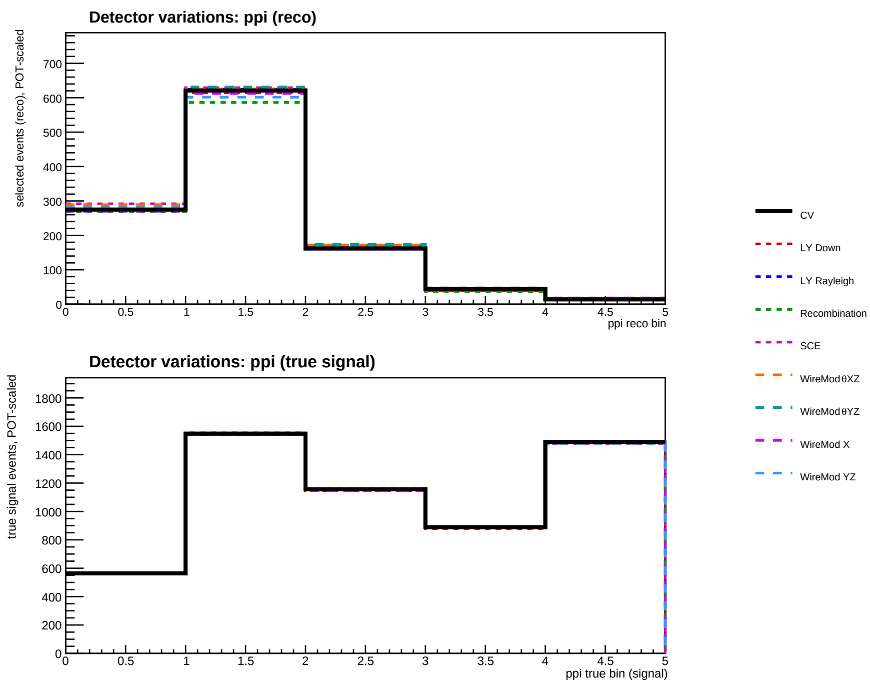

Figure 6: Detector variations, p_π.

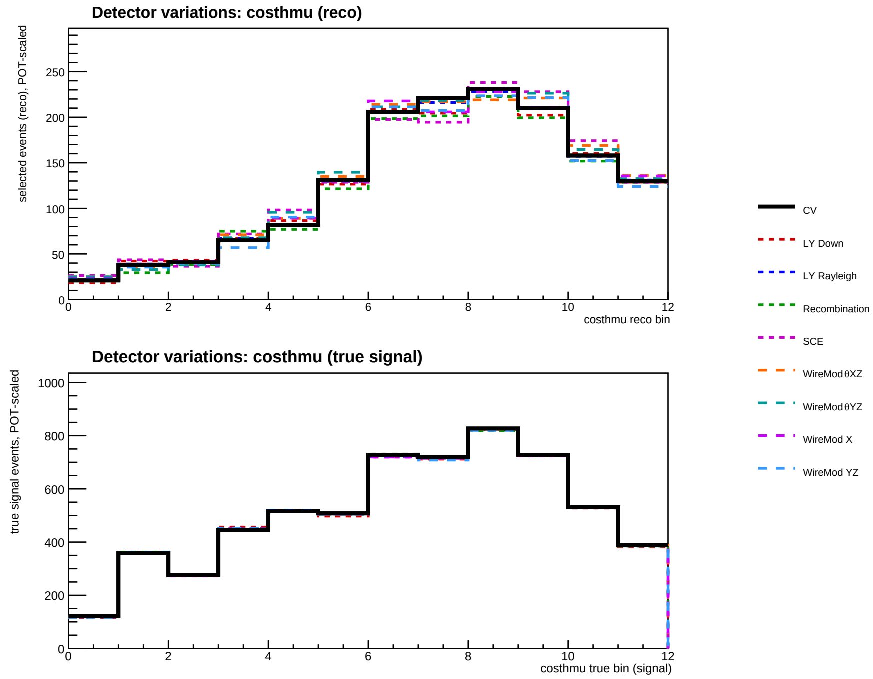

Figure 7: Detector variations, $\cos\theta_{\mu}$.

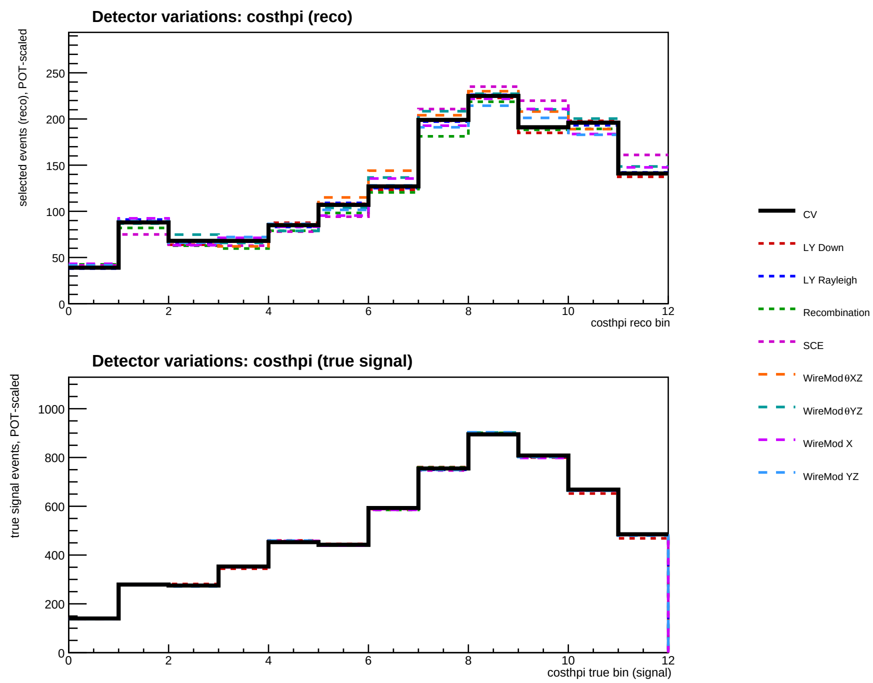

Figure 8: Detector variations, $\cos\theta_{\pi}$.

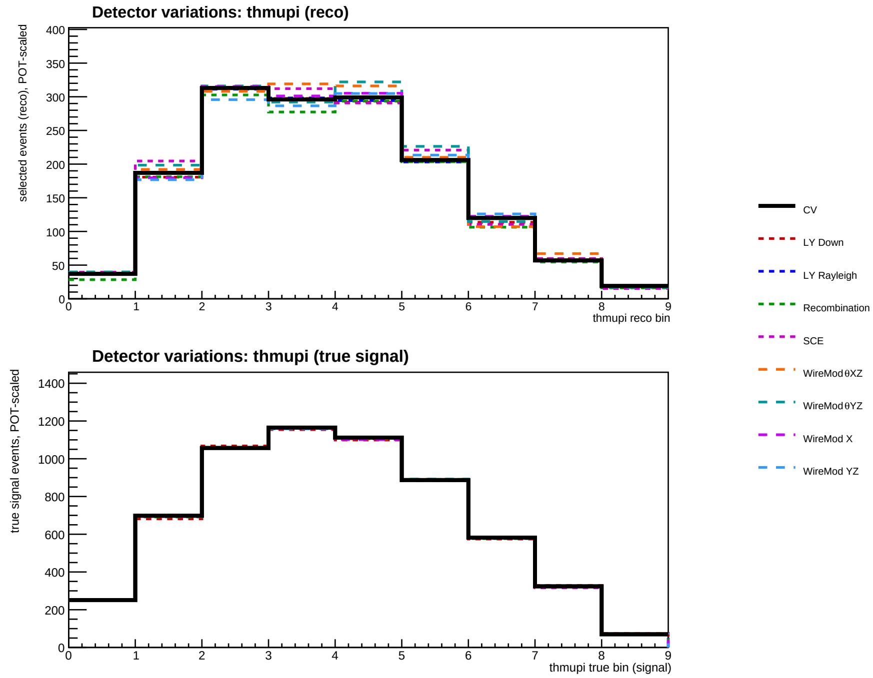

Figure 9: Detector variations, $\theta_{\mu\pi}$.

5.1 Unfolding

Wiener-SVD with the filter enabled and second-derivative regularisation (`Unfold WienerSVD 1 second-deriv`), chosen over D'Agostini after a variant study — it suppresses the bin-to-bin oscillation arising from sparse Run 1 statistics. D'Agostini with 4 iterations remains available as a cross-check via `configs/ccpi_xsec_config_numi_range_dagost.txt`.

As in the BNB analysis, the unfolded result must be compared to predictions smeared by the additional smearing matrix A_C.

6 Fake-Data Tests

All results in this section use the corrected flux (§2.4) and are drawn from a full re-run of `univmake` + `Unfolder/UnfolderNuMI` on all five observables. Predictions are compared in the space of the unfolded result, i.e. after the additional smearing matrix A_C is applied (§5.1).

6.1 GENIE closure test

The fake data is the detector-variation central-value GENIE sample (§2.2), statistically independent of the numuMC overlay that builds the response matrix. Treating it as data, subtracting the scaled beam-off sample and unfolding should recover its own true signal distribution. The closure is quantified by the χ^2 between the unfolded result and the A_C-smeared fake-data truth, using the full covariance:

Observable	χ^2 / ndf	p-value
p_μ	7.23 / 22	0.999
$\cos\theta_\mu$	4.40 / 12	0.975
$\cos\theta_\pi$	3.32 / 12	0.993
p_π	0.26 / 5	0.998
$\theta_{\mu\pi}$	2.26 / 9	0.987

All five observables close with $p > 0.97$: the unfolding machinery recovers the fake-data truth. (The very high p-values reflect the large Run-1 covariance, not overfitting.) This is a cleaner closure than the NuWro overlay could provide, since data and response now share the same generator model.

6.2 Fake-data normalisation offset

The unfolded fake data sits ~1.29× above the framework GENIE tune (the numuMC CV prediction), even though both are the same GENIE G18_10a tune. This was traced to a **flat** difference in events-per-POT between the detVar-CV sample and the numuMC overlay: the ratio is 1.293 integrated and 1.294 in the dominant catch-all true bin, i.e. present across *all* event types, with the signal-bin shape agreeing within Poisson statistics. It is therefore a POT / flux bookkeeping difference from using the Run-3b detVar-CV sample as Run-1 fake data — consistent with the framework's own caveat that detVar samples exist only for Run 3b and are applied globally (`SystematicsCalculator.cxx`) — not a physics or generator difference and not a framework bug.

The familiar "data ≈ 2× above the generators" then decomposes as **1.29× (fake-data normalisation) × ~1.3× (A_C smearing)**. It can be removed by entering the fake data with an effective POT of $7.63 \times 10^{20} \times 1.293 = 9.87 \times 10^{20}$ (and rebuilding the universes); this is documented but left uncorrected in `configs/file_properties_numi.txt`, since it does not affect the closure and vanishes entirely once the real beam-on sample replaces the fake data.

6.3 Generator comparison

Standalone predictions for the same signal definition and phase space were generated on the corrected "newg4" flux and overlaid on every unfolded plot. Each generator was run in the SL7 container environment:

Generator	Version / tune	CC1π total, $\cos\theta_\mu$ integral [10^{-38} cm²/Ar]
GENIE	3.04, tune G18_10a_02_11a	0.82
GiBUU	local release build, FSI on	1.13
NuWro	21.09	1.25
NEUT	local build, MPV-3 flux×σ sampling	1.32

The four span 0.82–1.32 (GENIE lowest, NEUT highest) — a healthy generator spread. Predictions are combined from ν_μ and $\bar{\nu}_\mu$ runs with the authoritative flux fractions ($\Phi_{\nu\mu} = 4.43515 \times 10^{-10}$, $\Phi_{\bar{\nu}\mu} = 2.37644 \times 10^{-10}$) and converted to per-nucleus 10^{-38} cm²/Ar (× A = 40). Two per-generator normalisation conventions had to be handled explicitly: NEUT's

`Totcrs` is per **nucleon** (not per nucleus), and GiBUU's `perweight` is already in 10^{-38} cm² units; getting these wrong produced 40× and 10^{38}× errors respectively before the fix.

The five comparison figures (`unfold_output/plot_{costhetamu,costhetapi,pmu, ppi,thetamupi}_0.pdf`) show, per observable: the unfolded fake data with its uncertainty, the four generators and the MicroBooNE tune (all A_C-smeared), the fake-data truth curve, and a data/MC ratio panel.

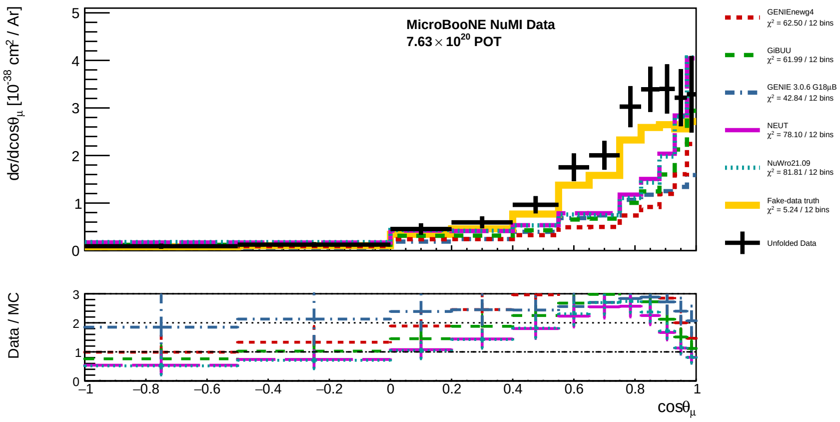

Figure 10: Unfolded $\mathrm{d}\sigma/\mathrm{d}\cos\theta_\mu$ (fake data), four generators + tune + fake-data truth, with data/MC ratio panel.

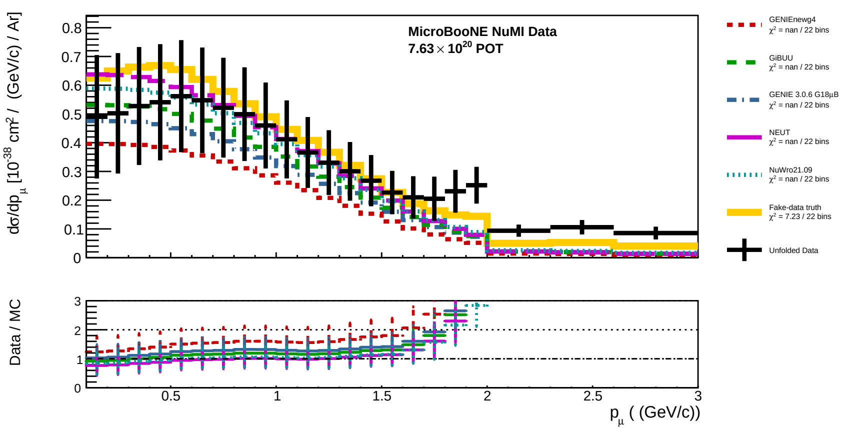

Figure 11: Unfolded $\mathrm{d}\sigma/\mathrm{d}p_\mu$.

6.4 NuWro overlay vs standalone NuWro (shape study)

The NuWro overlay previously used as fake data was compared bin-by-bin against a standalone NuWro 21.09 run on the same flux. Their cos θ_μ distributions differ in **shape**, not just normalisation: the overlay/standalone ratio swings from 0.64 (backward) to 1.88 (mid-angle) and back to 0.65 (forward), while the integral ratio is only ~1.1. The cause is *not* the flux — the two samples have identical signal-weighted mean neutrino energies (1.93 vs 1.88 GeV, 3%) — but a NuWro version/configuration difference in the muon angular distribution at

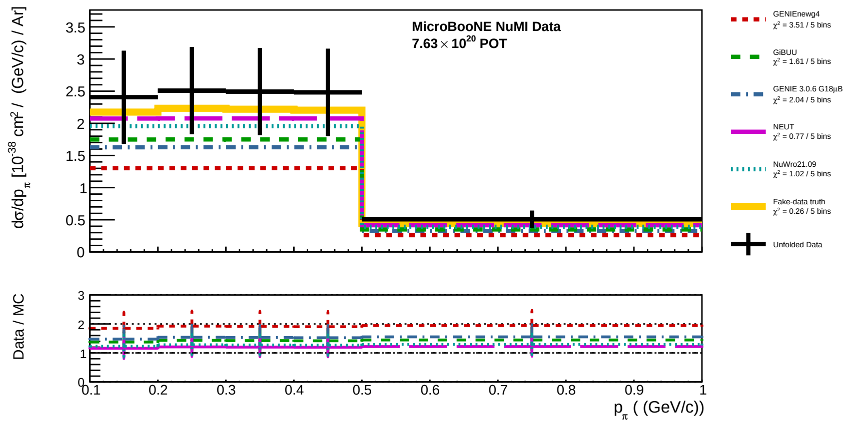

Figure 12: Unfolded $\mathrm{d}\sigma/\mathrm{d}p_\pi$.

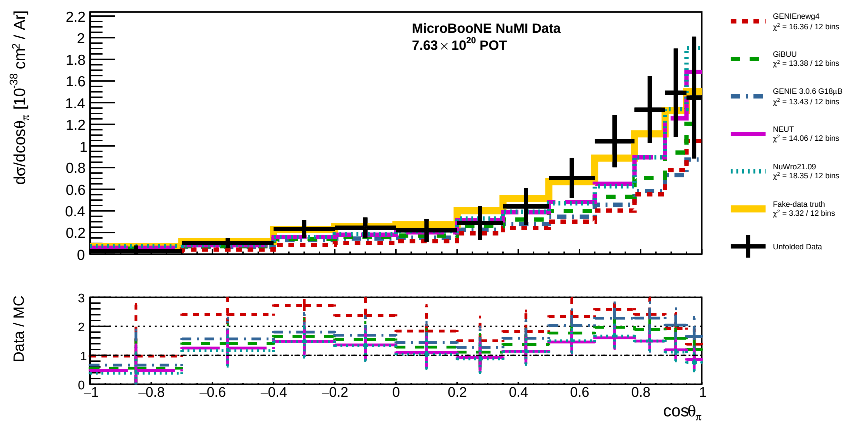

Figure 13: Unfolded $\mathrm{d}\sigma/\mathrm{d}\cos\theta_\pi$.

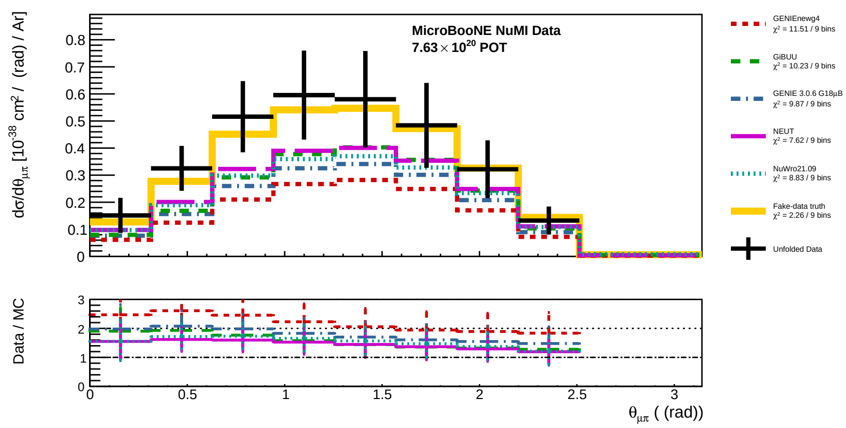

Figure 14: Unfolded $\mathrm{d}\sigma/\mathrm{d}\theta_{\mu\pi}$.

fixed energy. This is why the NuWro overlay was retired in favour of the independent-GENIE fake data (§2.2), and it is a reminder that an overlay fake sample is not a clean stand-in for an external generator prediction.

6.5 Detector-variation closure and the total cross section

Each of the eight detector variations was used in turn as the fake data — a full `univmake` rebuild with that variation in the `onBNB` slot (not a histogram swap, which was found to leave the measured reco unchanged), so the unfolded result genuinely responds to each variation. The total cross section (cos θ_μ integral) and the closure χ^2 are:

Variation	**Unfolded σ**	**True σ (raw)**	**Closure χ^2/12 (p)**
CV (ref)	1.289	0.975	4.40 (0.975)
LY Down	1.267	0.964	4.29 (0.978)
LY Rayleigh	1.280	0.972	4.07 (0.982)
Recombination	1.215	0.969	4.28 (0.978)
SCE	1.284	0.974	3.95 (0.984)
WireMod θXZ	1.293	0.978	3.33 (0.993)
WireMod θYZ	1.317	0.976	5.03 (0.957)
WireMod X	1.283	0.972	3.88 (0.985)
WireMod YZ	1.222	0.971	3.32 (0.993)

(σ in 10^{-38} cm^2/Ar.) Two results:

- **Every variation closes within uncertainty** ($\chi^2/12 \le 5.0$, p = 0.96–0.99): the unfolding recovers each variation’s truth as pseudo-data.
- **Detector systematic on the total.** The *unfolded* total spans 1.215–1.317, a **~±4 %** spread, while the *true* total barely moves (0.964–0.978, ±0.7 %) — as expected, since detector variations change reco, not truth. The extra spread in the unfolded total is the detector-response mismatch (the CV response applied to varied reco) and is a direct estimate of the detector contribution to the total-cross-section uncertainty.

The constant ~1.3× ratio of unfolded to *raw* true total is the Wiener-SVD additional-smearing matrix A_C (§5.1): the unfolded estimator lives in A_C-smeared space, so its central value sits above the raw truth by this fixed factor, which cancels in the closure χ^2. Per-variation plots are in `unfold_output/dv_rebuild/`.

6.6 Cross-section model excursions (M_A)

The sensitivity to the axial mass was probed in three steps.

CCQE axial (AxFFCCQEshape) ±σ — negligible. The framework already computes this unisim; its effect on the CC1π signal is **±0.08 %** (total), ≤0.5 % per bin. Expected: CC1π is resonance-dominated, so the CCQE axial form factor barely touches it. This is *not* the relevant axial mass for this channel.

Resonance axial mass M_A^RES ±σ (prediction). The standalone GENIE events were reweighted with `grwght1p -s MaCCRES` at ±1σ. The effect on the CC1π prediction is large and

asymmetric: **−5.9 % at −1σ, +32.5 % at +1σ** (consistent across all five observables). The asymmetry is intrinsic to GENIE's RES reweighting (the −1σ weights are bounded 0–1.15 while +1σ has a tail to ~2.8). The shift is **Q²-dependent**: strongest at backward/high-angle muons (+58 % at +1σ, cos θ_μ ≈ −1) and weakest forward (+17 %, cos θ_μ → 1), because backward muons carry higher Q² where the axial form factor's M_A dependence is strongest. The ±σ band is shown in the figure below. M_A^RES is the **dominant single-parameter cross-section systematic** for this measurement.

M_A^RES ±σ through the full unfolding. M_A^RES-shifted pseudo-data was built by reweighting the fake data's true signal by the per-true-bin factor f(t) and its measured reco by the migration-propagated g(r) = Σ_t mig(t,r) f(t) / Σ_t mig(t,r), then unfolded with the nominal response. The reweighted **truth** shifts by −6.5 % / +33.3 % (matching the prediction, confirming the reweight), and the **closure holds** for both variations (p = 0.95–0.98). The impact on the **unfolded total** is **−5.5 % / +28.2 %** — slightly smaller than the truth shift because the Wiener-SVD regularisation and A_C smearing damp the excursion by ~15 % (more so for the larger +1σ). So the analysis propagates the M_A^RES systematic correctly, with a mild regularisation-induced dilution.

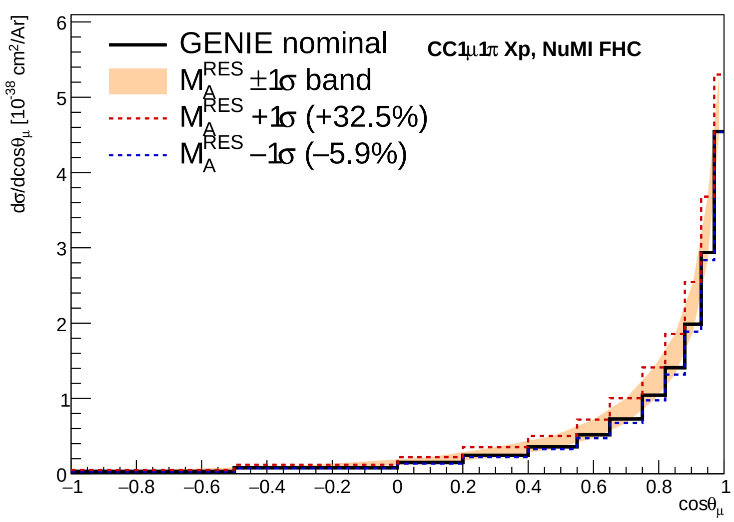

Figure 15: $M_A^{RES} \pm 1\sigma$ band on the GENIE CC1π $\cos\theta_\mu$ cross section (nominal black; $+1\sigma$ red, -1σ blue). The band is asymmetric and Q^2-dependent (largest at backward angles).

7 Results

Empty pending a real-data run. The fake-data machinery is now validated end-to-end (§6): the flux normalisation is corrected, the GENIE closure passes for all five observables, and four standalone generators are overlaid. What remains before this section can carry a **data** measurement is, in order:

1. Restore the real beam-on sample in `file_properties_numi.txt` (currently the GENIE detVar-CV fake data occupies the `onBNB` slot, §2.2).

2. Re-run the full chain from `ProcessNTuples` — the corrected momentum branches (§2.4) do not exist in the currently processed files.
3. Restore the NuMI beamline-geometry systematic (§4.6, §8.2); the uncertainty budget is under-estimated without it.

The units/normalisation questions (§8.1) and the flux scale (§8.5) are resolved.

This section should then contain, per observable and for the total: the unfolded cross section with the full uncertainty breakdown, and the generator comparison of §6.3 against real data rather than fake data.

8 Open Items

8.1 Cross-section units and bin-width normalisation. — RESOLVED. `conversion_factor()` divided by `1e39` in NuMI mode against `1e38` for BNB, and `Unfolder.C` omitted the bin-width division that `UnfolderNuMI.C` performs, so the two binaries disagreed on the same bin. Both are now fixed to the BNB convention (10^{-38} cm^2/Ar, differential): the NuMI factor is `1e38`, and `Unfolder.C` divides each bin by `conv_factor × bin_width × other_var_width` via `SliceHistogram::transform()`, which also converts the covariance. Verified on the costhmu NuWro config that `Unfolder` and `UnfolderNuMI` now agree bin-for-bin (ratio 1.0000 across all 12 bins). Relative to earlier output, every cross-section number is ~10× smaller and now differential.

8.2 Beamline geometry systematic. §4.6.

8.3 Anti-neutrino contamination. — CONSISTENT, by construction. The signal uses `|mc_nu_pdg| == 14`, admitting $\bar{\nu}_\mu$ CC events into signal and the efficiency denominator (the three sibling selections use an exact comparison). With the flux fix (§2.4) the normalising flux is now the **combined** $\nu_\mu + \bar{\nu}_\mu$ flux ($4.43515 \times 10^{-10} + 2.37644 \times 10^{-10}$), and the standalone generators are combined with the same fractions (§6.3). Signal, flux and predictions therefore all treat ν_μ and $\bar{\nu}_\mu$ together, so there is no longer a normalisation mismatch — the measured quantity is an effective flux-averaged $\nu_\mu+\bar{\nu}_\mu$ CC1π cross section. A ν_μ-only measurement would instead require restricting the signal to `mc_nu_pdg == 14` and using the ν_μ-only flux; that is a definition choice, not a bug.

8.4 Unused muon BDT. `tmvaReader_mu` is constructed and booked but never evaluated, and its input variables are never filled — suggesting an intended muon-PID cut was dropped.

8.5 Flux normalisation. — RESOLVED. The NuMI flux constant was ~3.48× too large (§2.4), making cross sections ~3.5× too small. Corrected to the flux author's authoritative value (6.81159×10^{-10} ν_μ/cm^2/POT, E > 60 MeV) in `FiducialVolume.hh` and confirmed four independent ways. A residual ~few-% active-volume-vs-fiducial-volume approximation is documented in that file.

8.6 Fake-data normalisation offset. — DOCUMENTED, uncorrected. The detVar-CV fake data sits a flat ~1.29× above the numuMC GENIE tune from a Run-3b-vs-Run-1 POT/flux bookkeeping difference (§6.2). It does not affect the closure and disappears when the real beam-on sample replaces the fake data; the POT correction to remove it (effective POT 9.87×10^{20}) is noted in `file_properties_numi.txt`.

Prepared as a companion to `BNBInternalNoteCC1pi_V0.91.pdf`*. Analysis code state: see git history on branch* `fix/systematics-warning`*. Fake-data figures:* `unfold_output/plot_*_0.pdf` *and the per-observable closure dumps* `unfold_output/closure_hists_ccpi_Run1_*_xsec.root`*. Standalone generator predictions and their build scripts:* `generator_predictions/newg4/`*.*